Appendix: Quantifying Hidden Exploitation

Online Supplementary Materials for Social Indicators Research Submission

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Preface

This appendix accompanies the main paper. Appendix A documents the codebook and how the composite risk index and the five binary exploitation indicators are constructed. Appendix B lists the survey-instrument questions used to elicit ego- and alter-based reports for each indicator. Appendix C describes the three-step bootstrap procedure used for the NSUM confidence intervals. Appendix D reports the sensitivity and robustness analyses summarised in the main paper.

All data and replication code are available at OSF. The composite risk index and all indicator variables are constructed in `R/data_processing/02-data_preparation.R` via `create_comparable_indicators()`.

A Variable Codebook and Indicator Definitions

This section documents the composite risk index construction, the five binary exploitation indicators used as the primary prevalence outcomes, and the key RDS variables.

A.1 Composite Risk Index

The composite risk index aggregates all eleven ILO indicators of forced labour into a single continuous measure (range 0–0.99). The index is a formative additive composite: an individual is exposed to exploitation if any of the indicators is present, and the indicators are constitutive rather than reflective of an underlying severity construct. Polarity is uniform across categories. Each of the eleven ILO indicator categories contributes equally to the composite (maximum 0.09 per category, so the total maximum possible score is $11 \times 0.09 = 0.99$). Within a category, the 0.09 is distributed evenly across the sub-questions used to code that category (0.09 for one sub-question, 0.045 each for two, 0.03 each for three, 0.0225 each for four). All eleven ILO indicators contribute equally to the composite. The medium/strong severity labels (following International Labour Organization (2024)) are used for reporting; they do not enter the composite's arithmetic.

Two further evidentiary markers — payment below the national living wage (Category 12) and NRM-referral history (Category 13) — are reported as separate binary flags and are **not** included in the composite, because their direct evidentiary status under UK policy makes them more useful when read independently.

Table A.1 lists each of the eleven ILO categories that enter the composite alongside the survey items used to code them, the coding rules, and the maximum weight per category. Categories 12 and 13 are shown at the bottom of the table for completeness but do not enter the composite score.

Table A.1: Composite Risk Index: ILO indicator categories, survey items, coding rules, and sub-weights. All eleven ILO indicator categories carry the same maximum category weight of 0.09; this 0.09 is split evenly across the sub-questions used to code the category (0.09 for one, 0.045 each for two, 0.03 each for three, 0.0225 each for four sub-questions). The total maximum composite score is therefore 11 $\times$ 0.09 = 0.99. The medium/strong severity labels in column 2 follow the ILO handbook on forced-labour surveys (International Labour Organization, 2024); the shorter ILO operational-indicators publications (International Labour Organization, 2012, 2025) list the eleven indicators without publishing this severity ranking, and the labels do not scale the composite weights. Categories 12 and 13 are reported as separate binary flags and are excluded from the composite.

#	ILO Indicator (severity)	Survey item(s)	Coding rule	Max weight
1	Abuse of vulnerability (medium)	Q29 (5a1): Forced to do work against will; Q51 (5d2-a): Unpaid tasks	Q29: 0/1/2 → 0.045; Q51: 0 → 0.045	0.09
2	Deception (medium)	Q45 (5c2): Forced/deceived/threatened into poor conditions	Q45: 0 → 0.09	0.09
3	Restriction of movement (medium)	Q32 (5a4): Could not leave premises; Q46 (5c3): Stay overnight permanently	Q32: 0/1/2 → 0.045; Q46: 2 or 3 → 0.045	0.09
4	Isolation (medium)	Q65: Have friends outside work	Q65: 1 (No) → 0.09	0.09
5	Physical and sexual violence (strong)	Q44 (5c1): Abuse/harassment/physical violence; Q44 (5c1-b): Sexual violence	Q44 (5c1): 0 → 0.045; Q44 (5c1-b): 0 → 0.045	0.09

#	ILO Indicator (severity)	Survey item(s)	Coding rule	Max weight
6	Intimidation and threats (strong)	Q47 (5c4): Employer threatened/intimidated	Q47: 0/1/2 → 0.09	0.09
7	Retention of identity documents (strong)	Q70 (5f2): Travel/identity docs withheld	Q70: 0/1/2 → 0.09	0.09
8	Withholding of wages (medium)	Q42 (5b7): Pay ever withheld	Q42: 0/1/2 → 0.09	0.09
9	Debt bondage (medium)	Q39 (5b4): Earnings to pay off debt to helper	Q39: 0 → 0.09	0.09
10	Abusive working and living conditions (medium)	Q48 (5c5): Verbally abused; Q63 (5d10): Lack of privacy; Q72 (5f4): Safe/healthy environment; Q76 (5f8): Access to food/water/sanitation	Q48: 0/1/2 → 0.0225; Q63: 0/1/2 → 0.0225; Q72: 2 or 3 → 0.0225; Q76: 1/2/3 → 0.0225	0.09
11	Excessive overtime (medium)	Q16 (3b): Weekly hours; Q61 (5d8): Weekly rest 24h; Q62 (5d9): Excessive overtime felt	Q16: 6 or 7 → 0.03; Q61: 2 or 3 → 0.03; Q62: 0/1/2 → 0.03	0.09
12	Payment below national living wage *(separate binary)*	Q36 (5b1): Hourly pay before tax	Q36: 0/1/2 (below £10.42/hr) → 1	Reported separately

#	ILO Indicator (severity)	Survey item(s)	Coding rule	Max weight
13	NRM-referral indicator *(separate binary)*	Q80 (5f12): Advised to enter National Referral Mechanism	Q80: 0 or 1 → 1	Reported separately

A.2 Five Binary Exploitation Indicators

The five binary indicators are the primary prevalence outcomes for both the ego-based RDS and alter-based NSUM analyses. All are binary (0/1). Multi-category RDS responses are recoded so that affirmative answers → 1, Never/no access → 0, and "Don't know"/"Prefer not to say" → NA. Composite RDS indicators combine sub-questions using logical OR (any affirmative response across sub-questions → 1). NSUM count variables (number of domestic workers known with the characteristic) are converted to binary ($> 0 \rightarrow 1$). Table A.2 summarises the mapping from each binary indicator to its ILO classification, the survey items that code it, and the coding rule applied.

Table A.2: Five Binary Exploitation Indicators: ILO classification, survey questions, and coding rules.

Binary indicator	ILO classification	Survey item(s)	Coding rule
Document withholding	Strong (Category 7)	Q70 (5f2): Travel/identity docs withheld	Q70: 0/1/2 → 1
Threats and abuse	Strong (Categories 5+6)	Q45 (5c2): Forced/deceived/threatened; Q47 (5c4): Intimidated; Q48 (5c5): Verbally abused	Any of: Q45=0; Q47=0/1/2; Q48=0/1/2 → 1
Pay issues	Medium (Categories 8+9)	Q39 (5b4): Debt bondage; Q42 (5b7): Pay withheld	Either Q39=0 or Q42=0/1/2 → 1

Binary indicator	ILO classification	Survey item(s)	Coding rule
Excessive hours	Medium (Category 11)	Q61 (5d8): Weekly rest < 24h; Q62 (5d9): Excessive overtime	Either Q61=2/3 or Q62=0/1/2 → 1
Limited access to help	Auxiliary	Q78 (5f10): Knows where to get help (reverse-coded)	Q78 indicating no access → 1

A.3 Key RDS Variables

Table A.3 lists the variables used by the RDS estimation pipeline (`R/analysis/03-rds_estimation.R` and downstream scripts). The `id`/`recruiter.id` pair defines the recruitment network on which all RDS weighting and successive-sampling procedures operate; `q13` is the reported network-degree variable used to weight respondents in inverse proportion to their reported degree; `wave` and `nationality_cluster` are stratifying variables used in diagnostic plots and subgroup sensitivity analyses. All variables are constructed in `R/data_processing/02-data_preparation.R`.

Table A.3: Key RDS Variables: variable name, type, definition, and survey source.

Variable	Type	Definition	Source
id	Numeric	Unique respondent identifier	—
recruiter.id	Numeric	id of recruiter; -1 = seed	—
referral_1 through referral_5	Numeric	ids of up to five recruits	—
q13 (2f)	Numeric (count)	Personal network size: number of other domestic workers known with contact details. Used as degree variable in RDS weighting and NSUM denominator.	Q13

Variable	Type	Definition	Source
1a_eligibility	Binary	Worked as domestic worker in past 12 months	Q1a
wave	Numeric (0, 1, 2, …)	Recruitment wave; 0 = seed; k = recruited by wave-(k−1) respondent	Derived
nationality_cluster	Categorical (4 levels)	Filipino, Latinx, British, Other	Q3, Q5

B Survey Questions and Comparable Ego/Alter Indicators

This section documents the full wording and response options for the ego-based (RDS) and alter-based (NSUM) survey questions underlying each of the five binary exploitation indicators. Ego-based questions ask respondents about their own experiences; alter-based questions ask what respondents observe in the domestic workers they know personally. This matched ego/alter pairing is the foundation of the dual-method design. All question numbers refer to the original survey instrument (*The Nature and Extent of Labour Exploitation among Domestic Workers in the UK*, administered via OnlineSurveys, University of Nottingham).

Table B.1 gives a summary overview; the subsections below provide the full question wording, response options, and coding rules for each indicator pair.

Table B.1: Comparable Ego/Alter Indicator Pairs — Summary. Confidence ratings reflect the closeness of the conceptual match between the ego (RDS) and alter (NSUM) question framings.

Indicator	Ego (RDS) question(s)	Alter (NSUM) question	Confidence
Document withholding	Q70 (5f2)	Q71 (5f3)	Highest
Pay/debt issues	Q39 (5b4) + Q42 (5b7)	Q43 (5b8)	High
Threats and abuse	Q45 (5c2) + Q47 (5c4) + Q48 (5c5)	Q49 (5c6)	High
Excessive hours	Q61 (5d8) + Q62 (5d9)	Q64 (5d11)	Medium
Limited access to help	Q78 (5f10) reverse-coded	Q79 (5f11)	Lower

B.1 Document Withholding

Ego-based — Q70 (5f2):

> *Has your employer, or their agent, withheld from you your travel and identity documentation?*
>
> Response options: Yes, always / Often / Sometimes / Never / I don't know

Alter-based — Q71 (5f3):

> *Do you know personally of any other domestic workers who do not have access to their own travel and identity documents? If so, how many? (Please indicate the number, or state '0' if this does not apply to anyone you know.)*
>
> Response: Integer count

Coding. Ego: Q70 {Yes always, Often, Sometimes} $\rightarrow$ 1; {Never, I don't know} $\rightarrow$ 0; missing $\rightarrow$ NA. Alter: Q71 $> 0 \rightarrow 1$; Q71 $= 0 \rightarrow 0$; missing $\rightarrow$ NA.

Confidence: Highest. The ego and alter questions are direct mirror images of the same exploitation behaviour. The only difference is perspective (self vs. others), not concept or scope.

B.2 Pay/Debt Issues

Ego-based — Q39 (5b4):

> *Do you have to use some or all of what you earn to pay off a debt to someone who helped you find work?*
>
> Response options: Yes / No / I don't know

Ego-based — Q42 (5b7):

> *Has your pay ever been withheld by your employer?*
>
> Response options: Yes, always / Often / Sometimes / Never / I don't know

Alter-based — Q43 (5b8):

Do you know personally of any other domestic workers who have experienced problems with debt or pay? If so, how many? (Please indicate the number, or state '0' if this does not apply to anyone you know.)

Response: Integer count

Coding. Ego: Q39 = Yes $\rightarrow$ 1; {No, I don't know} $\rightarrow$ 0 (debt bondage component). Q42 {Yes always, Often, Sometimes} $\rightarrow$ 1; {Never, I don't know} $\rightarrow$ 0 (pay withheld component). Composite `pay_issues_rds`: logical OR — Q39 = 1 OR Q42 = 1 $\rightarrow$ 1; both NA $\rightarrow$ NA. Alter: Q43 $> 0 \rightarrow 1$; Q43 $= 0 \rightarrow 0$; missing $\rightarrow$ NA.

Confidence: High. The alter question (Q43) covers both debt and pay problems, matching the combined ego indicator. Some conceptual mismatch remains because the alter question aggregates two distinct ego concepts into one.

B.3 Threats and Abuse

Ego-based — Q45 (5c2):

Have you been forced, deceived or threatened into poor working conditions?

Response options: Yes / No / Prefer not to say

Ego-based — Q47 (5c4):

Has your employer ever threatened or intimidated you?

Response options: Yes / Often / Sometimes / Never / I don't know

Ego-based — Q48 (5c5):

Has your employer ever verbally abused you (e.g. shouted, insulted, tried to humiliate you, or called you names, etc.)?

Response options: Yes, Often / Yes, Sometimes / Maybe once or twice / Never / I don't know

Alter-based — Q49 (5c6):

Do you know personally of any other domestic workers that have had any experiences related to the use of threat or force? If so, how many? (Please indicate the number, or state '0' if this does not apply to anyone you know.)

Response: Integer count

Coding. Ego: Q45 = Yes $\rightarrow$ 1; {No, Prefer not to say} $\rightarrow$ 0. Q47 {Yes, Often, Sometimes} $\rightarrow$ 1; {Never, I don't know} $\rightarrow$ 0. Q48 {Yes Often, Yes Sometimes, Maybe once or twice} $\rightarrow$ 1; {Never, I don't know} $\rightarrow$ 0. Composite `threats_abuse_rds`: logical OR — any of Q45/Q47/Q48 = 1 $\rightarrow$ 1; all three NA $\rightarrow$ NA. Alter: Q49 $> 0 \rightarrow 1$; Q49 $= 0 \rightarrow 0$; missing $\rightarrow$ NA.

Confidence: High. Q49 captures the combined concept of threat and force, aligning with the three ego sub-questions. The ego composite is slightly broader (it includes verbal abuse via Q48), so some scope mismatch remains at the margin.

B.4 Excessive Working Hours

Ego-based — Q61 (5d8):

> *Does your weekly rest last longer than 24 hours consecutively?*
>
> Response options: Yes, always / Often / Sometimes / Never / I don't know

Ego-based — Q62 (5d9):

> *How often have you worked overtime that you felt was excessive?*
>
> Response options: Always / Often / Sometimes / Never / I don't know

Alter-based — Q64 (5d11):

> *Do you know personally of any other domestic workers who have experienced any of the above related to labour rights? (Including working excessive overtime, a failure to obtain daily or weekly breaks or annual leave.) If so, how many? (Please indicate the number, or state '0' if this does not apply to anyone you know.)*
>
> Response: Integer count

Coding. Ego: Q61 {Sometimes, Never} $\rightarrow$ 1 (indicates *inadequate* rest, i.e., rest rarely or never exceeds 24 h); {Yes always, Often, I don't know} $\rightarrow$ 0. Q62 {Always, Often, Sometimes} $\rightarrow$ 1; {Never, I don't know} $\rightarrow$ 0. Composite `excessive_hours_rds`: logical OR $\rightarrow$ 1 if either condition present; both NA $\rightarrow$ NA. Alter: Q64 $> 0 \rightarrow 1$; Q64 $= 0 \rightarrow 0$; missing $\rightarrow$ NA.

Confidence: Medium. Q64 explicitly includes *annual leave* ("a failure to obtain … annual leave") which is not captured by Q61 or Q62 on the ego side. This scope mismatch is the largest of the five indicator pairs and is acknowledged as a limitation of the comparable-indicator strategy.

B.5 Limited Access to Help

Ego-based — Q78 (5f10):

> *Do you know who might help you if you are not being properly paid or properly treated?*
>
> Response options: Yes / No

Alter-based — Q79 (5f11):

> *Do you know personally of any other domestic workers who do NOT know where to go for help? If so, how many? (Please indicate the number, or state '0' if this does not apply to anyone you know.)*
>
> Response: Integer count

Coding. Ego: Q78 is reverse-coded — Q78 = No → 1 (respondent lacks access to help); Q78 = Yes → 0; missing → NA. Alter: Q79 > 0 → 1; Q79 = 0 → 0; missing → NA.

Confidence: Lower. Q78 directly reports the respondent's own knowledge of available support, while Q79 asks the respondent to assess *others'* knowledge. The reverse-coding of Q78 aligns the polarity (both = 1 signals vulnerability), but the alter question requires the respondent to perceive and report on peers' states of knowledge — a different and more demanding cognitive task than self-report. The conceptual match is therefore less direct than for the other four indicators.

B.6 General Data-Processing Notes

All five indicators are binary (0/1) after recoding. RDS multi-category responses are recoded so that affirmative responses → 1, and negative/unknown responses → 0 or NA as specified above. Composite RDS indicators (pay/debt issues, threats and abuse, excessive hours) apply logical OR: any single affirmative sub-response → indicator = 1; only if *all* sub-questions are missing does the composite take the value NA. NSUM count variables are converted to binary: count > 0 → 1; count = 0 → 0. These constructions are implemented in `create_comparable_indicators()` in `R/data_processing/02-data_preparation.R`, following the comparable-indicator specification agreed between the authors in November 2024.

C Three-Step Bootstrap Procedure

This section documents the custom bootstrap procedure used to construct 95% confidence intervals for all NSUM (MBSU) prevalence estimates. The procedure preserves the RDS network topology while allowing standard resampling-based uncertainty quantification.

Step 1 — Resampling. We resample from the observed RDS sample using a neighbourhood bootstrap. In the neighbourhood bootstrap, each respondent is resampled along with their immediate network neighbourhood (recruiter and recruits), preserving the ego-network topology and recruitment-tie structure following Yauck & Moodie (2022). Let $S^{(b)}$ denote the bootstrap replicate for iteration b. This resampling scheme is chosen over simple node-level resampling because it respects the dependent structure introduced by respondent-driven recruitment; it is the default bootstrap in the project's NSUM estimation pipeline.

Step 2 — Reweighting. For each bootstrap replicate b, RDS inclusion weights are recomputed from the replicate's degree distribution. Two weighting schemes are implemented and shown to produce equivalent results (Table D.6):

- *Volz–Heckathorn (RDS-II) weights:* $w_i^{(b)} \propto 1/d_i^{(b)}$, where $d_i^{(b)}$ is respondent i's network degree in replicate b.
- *Successive Sampling (SS) weights:* incorporate the sampling fraction and frame size N_F, computed numerically via the `RDS` package's successive-sampling estimator. At our sample sizes the SS and VH weights produce identical prevalence estimates (Table D.6).

Step 3 — NSUM estimation and aggregation. The MBSU estimator is applied to each bootstrap replicate with the recomputed weights. The MBSU prevalence formula (Feehan & Salganik, 2016) is:

$$\hat{p}_H^{(b)} = \frac{\sum_i w_i^{(b)}\, y_{iH}^{(b)}}{\sum_i w_i^{(b)}\, d_i^{(b)}}$$

where $y_{iH}^{(b)}$ is respondent i's binary indicator of knowing at least one member of hidden group H, and $d_i^{(b)}$ is their network degree. For the adjusted estimate, the basic prevalence is multiplied by $1/(\delta \cdot \tau)$:

$$\hat{p}_{H,\mathrm{adj}}^{(b)} = \hat{p}_H^{(b)} \cdot \frac{1}{\delta} \cdot \frac{1}{\tau}$$

Bootstrap aggregation uses $B = 500$ replicates. The 95% confidence interval is the empirical 2.5th–97.5th percentile interval across the B bootstrap prevalence estimates. Point estimates are computed from the full observed sample (not the bootstrap mean). Results are shown to be invariant to the choice of bootstrap resampling method (chain, neighbourhood, successive sampling, tree) in Table D.5.

Additional resampling options available in the pipeline but not used in the main analysis: Tree Bootstrap (sample entire recruitment trees with replacement), Chain Bootstrap (sample chains or subchains), and a Successive Sampling Bootstrap that mirrors the SS weight construction. These alternatives are documented in the project code (`R/analysis/NSUM/`) and their equivalence to the neighbourhood bootstrap is confirmed in Table D.5.

D Sensitivity Analyses

D.1 Population Size and Invariance of Prevalence

All prevalence estimators used in this paper are either algebraically free of the assumed total population size N, or depend on N only through finite-population corrections that are negligible at our sample sizes.

RDS-I (Salganik–Heckathorn) and RDS-II (Volz–Heckathorn): The prevalence formula $\hat{p} = \sum_i w_i y_i / \sum_i w_i$ contains no N; weights $w_i \propto 1/d_i$ are degree-based only.

MBSU NSUM: The prevalence formula $\hat{p}_H = \bar{y}_{FH}/\bar{d}_{FF}$ is a ratio of two weighted means and contains no N. Only the absolute count estimate $\hat{N}_H = \hat{p}_H \times N_F$ depends on N_F; prevalence does not.

RDS-SS and MA-RDS: These estimators formally depend on N through finite-population corrections. At $n = 85$ our sampling fraction is:

- $n/N = 0.17\%$ at $N = 50{,}000$
- $n/N = 0.085\%$ at $N = 100{,}000$
- $n/N = 0.009\%$ at $N = 980{,}000$
- $n/N = 0.005\%$ at $N = 1{,}740{,}000$

All values are well below the conventional 5% threshold at which finite-population corrections become non-negligible. Headline estimates use $N = 980{,}000$ (EU baseline for UK domestic workers); MA-RDS Bayesian analysis uses $N = 50{,}000$ for computational tractability. Cached estimates across all four N values agree to within bootstrap noise. Full empirical confirmation across population sizes is reported in Table D.7.

D.2 Supporting Estimates: Bayesian Model-Assisted RDS (MA-RDS)

Bayesian Model-Assisted (MA-RDS) posterior estimates follow Gile & Handcock (2015), implemented via the `sspse` package with $N = 50{,}000$ and `seed.selection = 'degree'`. Posterior means range from 34.1% (document withholding) to 84.0% (excessive hours), reproducing the RDS-SS indicator ordering (Table D.1). The MA-RDS posterior means are insensitive to both the choice of seed-selection method (degree, random, sample) and the assumed population size N: all combinations of the two produce identical posterior means (Table D.2), and the corresponding posterior distributions for document withholding under three seed-selection methods are visually indistinguishable (Figure D.1). Because the MA Bayesian estimator does not converge to a useful posterior on continuous outcomes at $n = 85$, MA-RDS is reported only for the five binary indicators, not for the composite risk index.

Table D.1: Bayesian MA-RDS Estimates for Five Binary Indicators, N = 50,000, seed.selection = 'degree'. Posterior mean (95% credible interval). Ordering of indicators preserved vs. RDS-SS estimates in Table 5 of the main text.

Indicator	MA-RDS posterior mean (95% CI)
Document withholding	34.1% (16.1–52.1%)
Pay issues	49.3% (35.2–63.4%)
Threats and abuse	57.1% (34.6–79.6%)
Limited access to help	58.2% (46.0–70.4%)
Excessive working hours	84.0% (72.3–95.8%)

Table D.2: MA-RDS Sensitivity Analysis: Posterior Estimates across Population Sizes and Seed-Selection Methods. All combinations produce the same posterior means, confirming N-invariance and seed-method robustness. Posterior mean shown; 95% CI available on request.

Indicator	N = 50,000 degree	N = 50,000 random	N = 50,000 sample	N = 980,000 degree	N = 980,000 random	N = 980,000 sample	N = 1,740,000 degree	N = 1,740,000 random	N = 1,740,000 sample
Document withholding	34.1%	34.1%	34.1%	34.1%	34.1%	34.1%	34.1%	34.1%	34.1%

Indicator	N = 50,000 degree	N = 50,000 random	N = 50,000 sample	N = 980,000 degree	N = 980,000 random	N = 980,000 sample	N = 1,740,000 degree	N = 1,740,000 random	N = 1,740,000 sample
Pay issues	49.3%	49.3%	49.3%	49.3%	49.3%	49.3%	49.3%	49.3%	49.3%
Threats and abuse	57.1%	57.1%	57.1%	57.1%	57.1%	57.1%	57.1%	57.1%	57.1%
Limited access to help	58.2%	58.2%	58.2%	58.2%	58.2%	58.2%	58.2%	58.2%	58.2%
Excessive working hours	84.0%	84.0%	84.0%	84.0%	84.0%	84.0%	84.0%	84.0%	84.0%

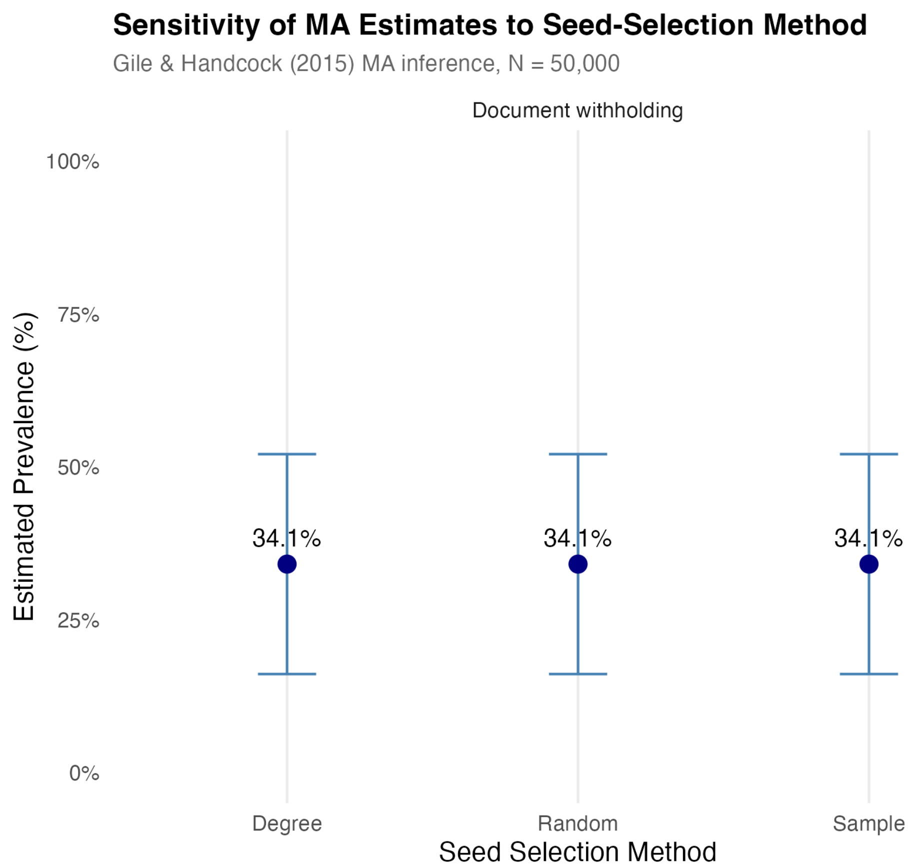

Figure D.1: Model-Assisted Seed-Selection Sensitivity. Bayesian MA posterior estimates and 95% credible intervals for document withholding under three seed-selection methods (degree, random, sample) at N = 50,000. All three methods converge to the same posterior, supporting model robustness to initialisation. Source: Authors' own work.

D.3 Supporting Estimates: RDS-I and RDS-SS Comparison

Table D.3 reports prevalence estimates and neighbourhood-bootstrap 95% confidence intervals for both the RDS-I (Salganik–Heckathorn) and RDS-SS (Gile's Successive Sampling) estimators across all four assumed population sizes. RDS-I runs systematically higher than RDS-SS by 5–15 percentage points, consistent with the standard finding that RDS-I is sensitive to seed dependence and performs poorly at small sample sizes and shallow chain depths. The gap between RDS-I and RDS-SS is larger for indicators where seed-cluster homophily is strongest. Estimates are essentially invariant across population sizes (confirming Section D.1). The preferred estimator for binary outcomes is RDS-SS; see main text Section 3.2. Figure D.2 visualises the RDS-I / RDS-SS gap across the five indicators as paired point estimates with bootstrap CIs at N = 980,000, showing that RDS-I sits above RDS-SS for every indicator with the largest gap for document withholding and the smallest for limited access to help.

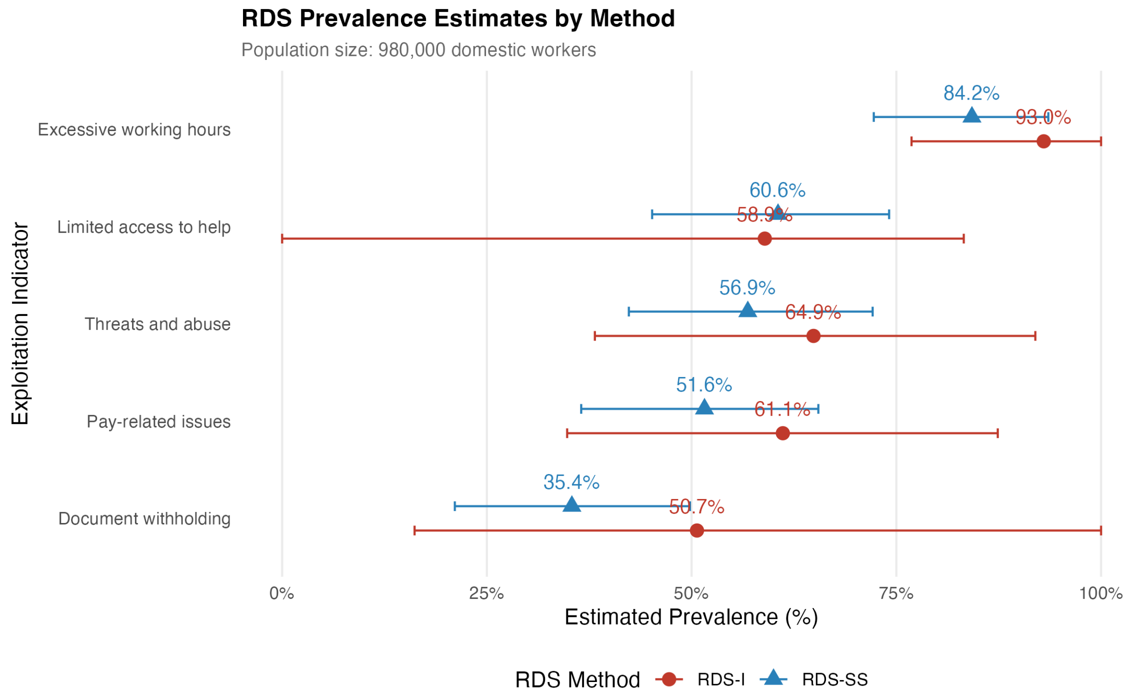

Figure D.2: RDS Prevalence Estimates by Method. Neighbourhood-bootstrap 95% confidence intervals for RDS-I (Salganik–Heckathorn) and RDS-SS (Gile's Successive Sampling) estimators for the five binary exploitation indicators, N = 980,000. RDS-I runs systematically higher than RDS-SS by 5–15 percentage points, consistent with seed dependence in the shallow-chain design. Source: Authors' own work.

Table D.3: Comprehensive RDS Estimates: Both Estimators across All Population Sizes. Neighbourhood-bootstrap 95% CI in parentheses. Estimates are invariant across N values; full table shows all four population sizes for completeness.

Indicator	Estimator	N = 50,000	N = 100,000	N = 980,000	N = 1,740,000
Document withholding	RDS-I	49.4% (25.3–56.3%)	49.4% (25.3–56.3%)	49.4% (25.3–56.3%)	49.4% (25.3–56.3%)
Document withholding	RDS-SS	34.2% (25.2–56.0%)	34.2% (25.2–56.0%)	34.2% (25.2–56.0%)	34.2% (25.2–56.0%)
Pay issues	RDS-I	59.3% (34.2–63.5%)	59.3% (34.2–63.5%)	59.3% (34.2–63.5%)	59.3% (34.2–63.5%)

Indicator	Estimator	N = 50,000	N = 100,000	N = 980,000	N = 1,740,000
Pay issues	RDS-SS	49.7% (34.8–63.4%)	49.7% (34.8–63.4%)	49.7% (34.8–63.4%)	49.7% (34.8–63.4%)
Threats and abuse	RDS-I	65.2% (40.5–72.1%)	65.2% (40.5–72.1%)	65.2% (40.5–72.1%)	65.2% (40.5–72.1%)
Threats and abuse	RDS-SS	57.2% (40.1–72.3%)	57.2% (40.1–72.3%)	57.2% (40.1–72.3%)	57.2% (40.1–72.3%)
Limited access to help	RDS-I	57.7% (55.1–80.3%)	57.7% (55.1–80.3%)	57.7% (55.1–80.3%)	57.7% (55.1–80.3%)
Limited access to help	RDS-SS	59.4% (54.7–80.2%)	59.4% (54.7–80.2%)	59.4% (54.7–80.2%)	59.4% (54.7–80.2%)
Excessive working hours	RDS-I	92.8% (71.4–92.9%)	92.8% (71.4–92.9%)	92.8% (71.4–92.9%)	92.8% (71.4–92.9%)
Excessive working hours	RDS-SS	83.7% (71.5–93.0%)	83.7% (71.5–93.0%)	83.7% (71.5–93.0%)	83.7% (71.5–93.0%)

D.4 Supporting Estimates: NSUM under Alternative Adjustment-Factor Tiers

For comparison with the discrete-tier NSUM literature, Table D.4 reports MBSU prevalence under three reference scenarios: Unadjusted ($\delta = 1.0$, $\tau = 1.0$), Moderate ($\delta = 0.9$, $\tau = 0.85$), and Conservative ($\delta = 0.8$, $\tau = 0.70$). These correspond to three points on the continuous sensitivity curve reported in Section D.5. The GNSUM Symmetric estimator produces results identical to MBSU No Adjustment. All estimates use $N_F = 980{,}000$ and neighbourhood-bootstrap 95% CIs ($B = 500$). Figure D.3 displays the MBSU Moderate estimates as a forest plot: the five indicators all cluster in the 1–3% range under the Moderate adjustment scenario, with excessive hours and pay issues at the top of the range and document withholding at the bottom.

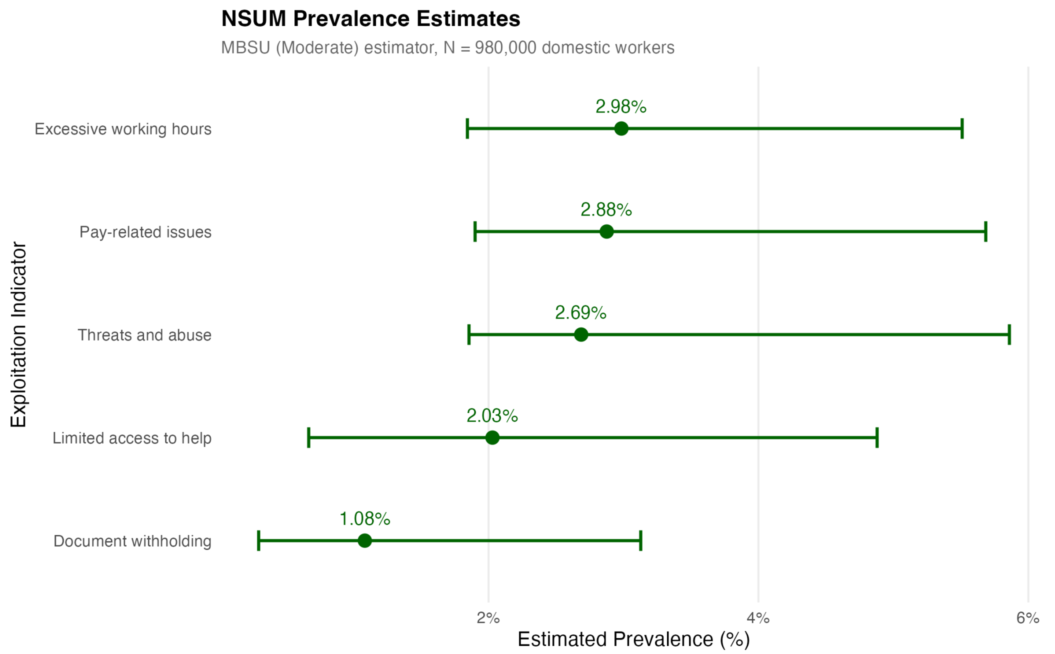

Figure D.3: NSUM (MBSU Moderate) Prevalence Estimates. Forest plot of MBSU Moderate ($\delta = 0.9$, $\tau = 0.85$) prevalence estimates with neighbourhood-bootstrap 95% confidence intervals for the five binary exploitation indicators, $N_F = 980{,}000$. Source: Authors' own work.

Table D.4: NSUM Prevalence under Alternative Estimators and Adjustment-Factor Scenarios, N = 980,000. Neighbourhood-bootstrap 95% CI in parentheses (B = 500). GNSUM Symmetric produces results identical to MBSU No Adjustment.

NSUM method	Indicator	Prevalence (95% CI)
MBSU No Adjustment (=1.0, τ=1.0)	Access to help	1.55% (0.51–3.73%)
MBSU No Adjustment (=1.0, τ=1.0)	Document withholding	0.83% (0.23–2.39%)
MBSU No Adjustment (=1.0, τ=1.0)	Excessive hours	2.28% (1.41–4.21%)
MBSU No Adjustment (=1.0, τ=1.0)	Pay issues	2.20% (1.45–4.35%)
MBSU No Adjustment (=1.0, τ=1.0)	Threats and abuse	2.06% (1.42–4.48%)
MBSU Moderate (=0.9, τ=0.85)	Access to help	2.03% (0.67–4.88%)

NSUM method	Indicator	Prevalence (95% CI)
MBSU Moderate (=0.9, τ=0.85)	Document withholding	1.08% (0.30–3.13%)
MBSU Moderate (=0.9, τ=0.85)	Excessive hours	2.98% (1.84–5.51%)
MBSU Moderate (=0.9, τ=0.85)	Pay issues	2.88% (1.90–5.68%)
MBSU Moderate (=0.9, τ=0.85)	Threats and abuse	2.69% (1.86–5.86%)
MBSU Conservative (=0.8, τ=0.70)	Access to help	2.77% (0.91–6.67%)
MBSU Conservative (=0.8, τ=0.70)	Document withholding	1.48% (0.40–4.27%)
MBSU Conservative (=0.8, τ=0.70)	Excessive hours	4.08% (2.52–7.53%)
MBSU Conservative (=0.8, τ=0.70)	Pay issues	3.93% (2.60–7.77%)
MBSU Conservative (=0.8, τ=0.70)	Threats and abuse	3.67% (2.53–8.00%)
GNSUM Symmetric	Access to help	1.55% (0.51–3.73%)
GNSUM Symmetric	Document withholding	0.83% (0.23–2.39%)
GNSUM Symmetric	Excessive hours	2.28% (1.41–4.21%)
GNSUM Symmetric	Pay issues	2.20% (1.45–4.35%)
GNSUM Symmetric	Threats and abuse	2.06% (1.42–4.48%)

D.5 Continuous-τ NSUM Sensitivity Sweep

The headline NSUM result in this paper is not the discrete-tier comparison of Section D.4 but rather the continuous sensitivity sweep: for each of the five exploitation indicators, MBSU prevalence is shown as a continuous function of the visibility/true-positive-rate adjustment parameter $\tau \in [0.40, 1.00]$, in increments of 0.05 (13 values), with δ fixed at 0.9 and $B = 500$ neighbourhood-bootstrap 95% confidence intervals at each τ value.

The three discrete tiers reported in Section D.4 correspond to three points on these continuous curves: Unadjusted at $\tau = 1.0$, Moderate at $\tau = 0.85$, and Conservative at $\tau = 0.70$. Prevalence increases monotonically as τ decreases (lower τ means lower assumed visibility, hence the estimator must scale up the raw alter-report count). The increase is modest across the plausible range $\tau \in [0.7, 1.0]$ and becomes substantial only at $\tau \leq 0.5$, which corresponds to an assumption that fewer than half of true exploitation cases are visible to the respondent's network — an extreme scenario. The continuous picture in Figure D.4 supersedes the discrete-tier comparison for substantive inference about how visibility assumptions affect estimates: it makes explicit that the paper's conclusions about indicator ordering and the RDS–NSUM visibility gradient are robust across all plausible τ values.

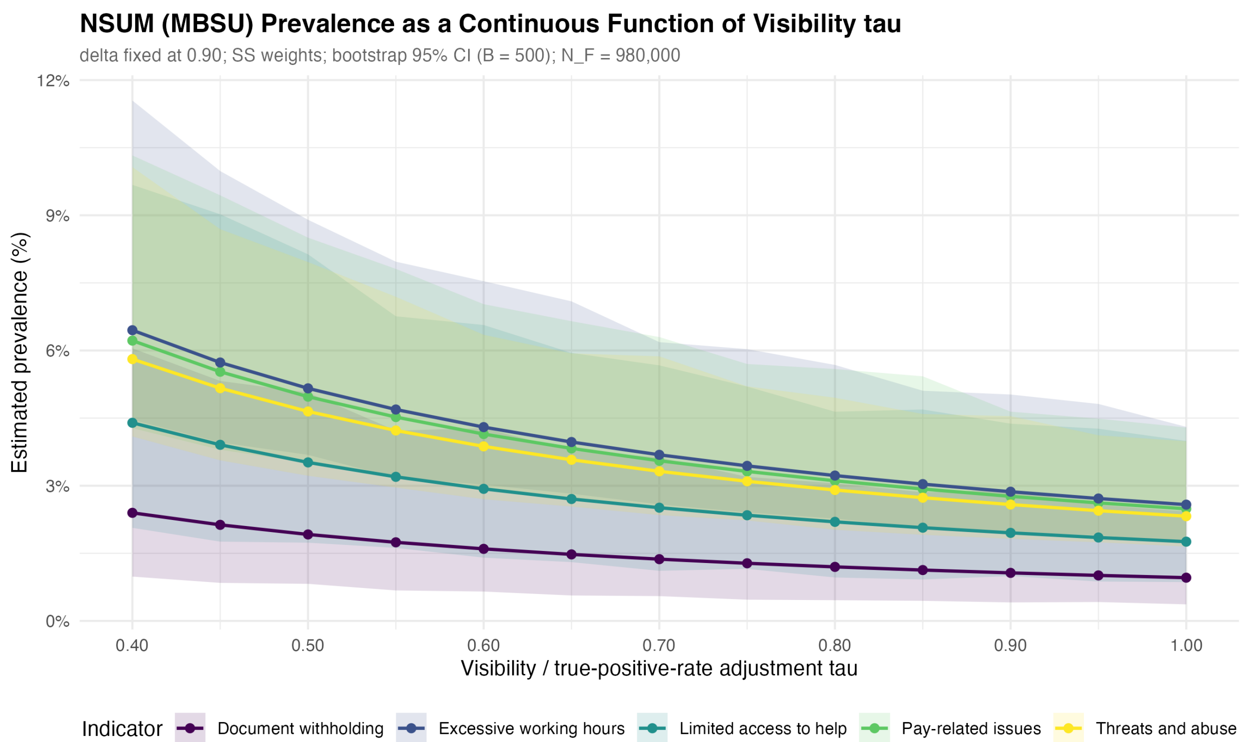

Figure D.4: NSUM (MBSU) Prevalence as a Continuous Function of Visibility Parameter τ. MBSU prevalence estimates for five exploitation indicators as a function of the visibility/true-positive-rate adjustment τ, with fixed at 0.9 and B = 500 neighbourhood-bootstrap 95% confidence intervals. Three discrete scenarios from the main text (Unadjusted: $\tau = 1.0$; Moderate: $\tau = 0.85$; Conservative: $\tau = 0.70$) correspond to three points on each curve. N_F = 980,000. Source: Authors' own work.

D.6 Robustness: Bootstrap Method and Weighting Scheme

Table D.5, Table D.6, and Table D.7 confirm that the headline NSUM estimates are robust to the choice of bootstrap resampling method, RDS weighting scheme, and assumed population size. In all cases the MBSU Moderate ($\delta = 0.9$, $\tau = 0.85$) point estimates are identical across variants; only confidence interval widths vary slightly across bootstrap methods, and the differences are substantively negligible.

Table D.5: Bootstrap Method Comparison. MBSU Moderate prevalence and neighbourhood-bootstrap 95% CI under four resampling methods. N = 980,000, SS weights. Point estimates identical across methods; CI widths vary within ±0.5 percentage points.

Bootstrap method	Indicator	Prevalence (95% CI)
Chain	Access to help	2.03% (0.93–4.73%)
Chain	Document withholding	1.08% (0.43–2.83%)
Chain	Excessive hours	2.98% (2.07–5.39%)
Chain	Pay issues	2.88% (1.99–5.08%)
Chain	Threats and abuse	2.69% (1.92–4.85%)
Neighbourhood (main analysis)	Access to help	2.03% (0.67–4.88%)
Neighbourhood (main analysis)	Document withholding	1.08% (0.30–3.13%)
Neighbourhood (main analysis)	Excessive hours	2.98% (1.84–5.51%)
Neighbourhood (main analysis)	Pay issues	2.88% (1.90–5.68%)
Neighbourhood (main analysis)	Threats and abuse	2.69% (1.86–5.86%)
Successive Sampling	Access to help	2.03% (0.99–4.66%)
Successive Sampling	Document withholding	1.08% (0.41–2.77%)
Successive Sampling	Excessive hours	2.98% (2.13–5.27%)
Successive Sampling	Pay issues	2.88% (2.06–4.94%)
Successive Sampling	Threats and abuse	2.69% (1.92–4.56%)
Tree	Access to help	2.03% (0.96–4.59%)
Tree	Document withholding	1.08% (0.45–2.71%)
Tree	Excessive hours	2.98% (2.09–5.40%)
Tree	Pay issues	2.88% (2.06–5.05%)
Tree	Threats and abuse	2.69% (1.91–4.70%)

Table D.6: Weighting Scheme Comparison. MBSU Moderate prevalence under Successive Sampling (SS) and Volz–Heckathorn (VH) weights, neighbourhood bootstrap, N = 980,000. SS and VH produce identical point estimates and confidence intervals.

Weight scheme	Indicator	Prevalence (95% CI)
SS (main analysis)	Access to help	2.03% (0.67–4.88%)
SS (main analysis)	Document withholding	1.08% (0.30–3.13%)
SS (main analysis)	Excessive hours	2.98% (1.84–5.51%)
SS (main analysis)	Pay issues	2.88% (1.90–5.68%)
SS (main analysis)	Threats and abuse	2.69% (1.86–5.86%)
VH (Volz–Heckathorn)	Access to help	2.03% (0.67–4.88%)
VH (Volz–Heckathorn)	Document withholding	1.08% (0.30–3.13%)
VH (Volz–Heckathorn)	Excessive hours	2.98% (1.84–5.51%)
VH (Volz–Heckathorn)	Pay issues	2.88% (1.90–5.68%)
VH (Volz–Heckathorn)	Threats and abuse	2.69% (1.86–5.86%)

Table D.7: Population Size Comparison. MBSU Moderate prevalence across four assumed domestic-worker population sizes. Neighbourhood bootstrap, SS weights. Prevalence is invariant across N values; absolute count estimates scale proportionally with N.

Indicator	N = 50,000	N = 100,000	N = 980,000	N = 1,740,000
Access to help	2.03%	2.03%	2.03%	2.03%
Document withholding	1.08%	1.08%	1.08%	1.08%
Excessive hours	2.98%	2.98%	2.98%	2.98%
Pay issues	2.88%	2.88%	2.88%	2.88%
Threats and abuse	2.69%	2.69%	2.69%	2.69%

D.7 Survey Incentive Design

The survey incentive structure followed the recommendations of Cobanoglu & Cobanoglu (2003), Brown et al. (2006), and Laguilles et al. (2011). A lottery entry into a free prize draw for £150 was offered to all respondents completing the questionnaire. Research by Gajic et al. (2012) suggests that a high-value lottery provides the most cost-effective incentive for survey participation in this type of study.

To minimise fraudulent responses, mobile phone numbers for each respondent and those they referred were collated, and each was called by one of the authors to verify respondent status as a domestic worker in the UK. This verification step is particularly important in an RDS study where respondents are recruited by peers, as it guards against chain contamination.

The ethics of survey incentives for hidden and potentially vulnerable populations have been carefully considered following Semaan et al. (2009), DeJong et al. (2009), and Singer & Bossarte (2006). Our incentive design follows the consensus recommendations of that literature: the incentive is proportionate and not coercive; participation is voluntary; and additional support resources were made available to respondents through the partner organisations (Kalayaan and the Latin American Women's Rights Service) who facilitated the data collection. Full ethical approval was obtained prior to fieldwork.

D.8 Leave-One-Out Chain Sensitivity

The main text notes that one chain (root seed #55, length 12) accounts for roughly a third of wave-2+ respondents, raising a question about whether the rank ordering of exploitation indicators is driven by that single chain. This section reports a leave-one-out (LOO) sensitivity analysis in which the chain rooted at seed #55 is entirely excluded, and RDS-SS point estimates and 95% bootstrap confidence intervals are recomputed on the reduced sample.

Chain identification. Chain membership is determined by breadth-first search (BFS) from the root node: starting with seed #55, all respondents whose `recruiter.id` traces back to seed #55 (directly or transitively) are collected. The BFS is implemented in `R/analysis/08c-loo_chain_analysis.R`. The chain contains 12 respondents — seed #55 plus 11 recruits spanning four subsequent recruitment waves — leaving a LOO analysis sample of $n = 73$. Table D.8 lists the twelve excluded respondents; eleven are Filipino and one is Latinx, and reported network degrees range from 0 to 30 (seed #55 is the highest-degree node in the chain, though not the highest-degree respondent in the whole sample).

Table D.8: Excluded chain (seed #55). Twelve respondents removed by the LOO analysis, showing recruitment wave (0 = seed), self-reported network degree (Q13), nationality cluster, and count of ILO-aligned binary indicators reported affirmatively (of five). Ordered by wave then by respondent id. Source: Authors' reconstruction from `data/processed/prepared_data.csv` via breadth-first search on `recruiter.id`.

Respondent id	Wave	Degree (Q13)	Nationality	Indicators affirmed (of 5)
55 (seed)	0	30	Filipino	1
41	1	5	Filipino	4
63	1	5	Filipino	4

Respondent id	Wave	Degree (Q13)	Nationality	Indicators affirmed (of 5)
6	2	10	Filipino	2
10	2	2	Filipino	3
42	2	10	Filipino	4
44	2	5	Filipino	4
12	3	0	Filipino	2
18	3	10	Latinx	1
90	3	1	Filipino	3
45	4	10	Filipino	1
64	4	10	Filipino	4

Re-estimation. For each of the five comparable indicators, RDS-SS estimates (Gile's Successive Sampling Horvitz-Thompson estimator) are computed on (a) the full sample ($n = 85$) and (b) the LOO sample ($n = 73$), using identical settings: $N_F = 980{,}000$, successive-sampling weights, neighbourhood-bootstrap 95% CIs ($B = 500$). The LOO `rds.data.frame` object is reconstructed with the same parameters as the main analysis (network size variable: `known_network_size`, max coupons: 5).

Result. Table D.9 reports the two sets of estimates side by side; Figure D.5 displays them as a paired forest plot. Point-estimate shifts between the full and LOO samples range from −3.2 to +4.5 percentage points across the five indicators, and every LOO 95% CI overlaps the corresponding full-sample CI. The top and bottom of the ordering are preserved (document withholding lowest and excessive hours highest in both samples), but the middle of the ordering changes: *threats and abuse* and *limited access to help* swap adjacent positions when seed #55's chain is removed (full-sample order: Doc < Pay < Threats < Access < Excessive; LOO order: Doc < Pay < Access < Threats < Excessive). Because the swap sits entirely inside overlapping confidence intervals for both indicators, it does not represent statistical evidence of a rank change; but it does mean the "rank order identical" claim in earlier drafts of this section overstates the robustness of the middle of the ordering. The LOO sensitivity confirms the qualitative substantive claim — that the top-and-bottom prevalence structure is not driven by seed #55's chain — while making explicit that adjacent-pair ordering in the middle of the range depends on which chains happen to enter the sample.

Table D.9: Full-sample vs Leave-One-Out RDS-SS Estimates. Preferred prevalence estimates for the five binary ILO-aligned indicators on the full sample ($n = 85$) and on the LOO sample with seed #55's chain excluded ($n = 73$). Neighbourhood-bootstrap 95% confidence intervals in parentheses. $N_F = 980{,}000$, SS weights, $B = 500$. All five full-vs-LOO CI pairs overlap. Source: `output/tables/loo_chain_sensitivity.csv`.

Indicator	Full sample (n = 85)	LOO sample (n = 73)	Shift (pp)
Document withholding	35.4% (20.3–48.1%)	33.4% (17.5–48.8%)	−2.0
Pay/debt issues	51.6% (37.2–65.9%)	55.3% (41.0–71.0%)	+3.8
Threats and abuse	56.9% (41.7–71.3%)	61.4% (46.5–76.7%)	+4.5
Excessive hours	84.2% (72.9–94.1%)	82.5% (69.7–93.7%)	−1.7
Limited access to help	60.6% (45.4–73.2%)	57.3% (41.3–72.6%)	−3.2

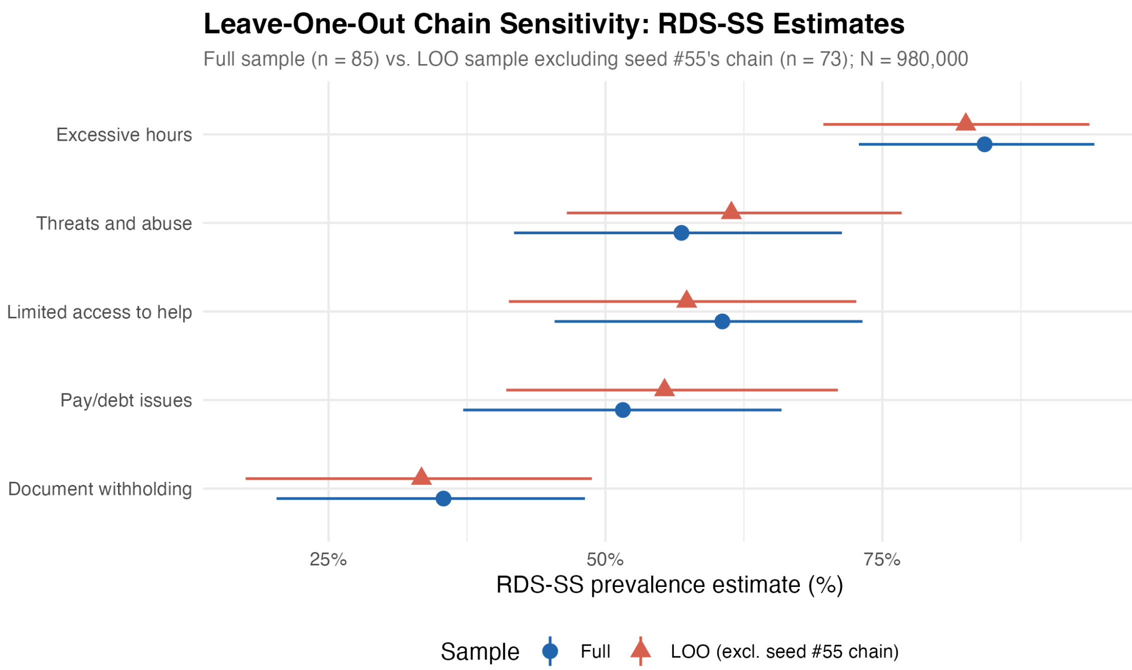

Figure D.5: Leave-One-Out Chain Sensitivity. RDS-SS point estimates and neighbourhood-bootstrap 95% confidence intervals for the five exploitation indicators in the full sample (n = 85) and in the LOO sample with seed #55's chain excluded (n = 73). N = 980,000. Source: Authors' own work.

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